Quiz MCQ – Final Assessment

Course: Full Stack AI B 5 SAT,SUN)
Dated:3rd March 2026
Name: Amna Ameer
CNIC No: 35201-7317984-4

🧮 NumPy — 5 MCQs

1) What is the primary advantage of using NumPy arrays over Python lists for numeric computation?

C. NumPy arrays are contiguous and support vectorized operations

2) What is the result shape of *np.sum(a, axis=0)* **where** *a* **is shape** *(3, 4)***?**

C. *(4,)*

3) Which statement about broadcasting is TRUE?

B. Trailing dimensions are compared and dimensions of size 1 are “stretched”

4) Which creates a *view* most often (not guaranteed copy)?

C. *a.ravel()*

5) What does *-1* **mean in** *a.reshape(-1, 2)***?**

C. Infer that dimension from the total size

🐍 Python – List — 5 MCQs

1) What’s the difference between *append* **and** *extend***?**

B. *append* adds an element; *extend* concatenates an iterable

2) What happens with *lst = [[0]] * 3; lst[0].append(1)***?**

B. All inner lists change

3) What's the result of *sorted([3, 1, 2])* **vs** *[3,1,2].sort()***?**

C. *sorted* returns new; *sort* modifies in place and returns *None*

4) What does *lst[::-1]* **do?**

C. Returns a reversed slice

5) Which removes by value (first occurrence)?

C. *remove()*

📦 Python – Tuples — 5 MCQs

1) Which creates a single-element tuple?

B. *(1,)*

2) What is TRUE about tuples?

B. Tuples are immutable but can contain mutable elements

3) What does *a, b = b, a* **do?**

A. Swaps values without temp variable

4) Which is valid tuple unpacking with a starred expression?

A. *a, *b, c = [1,2,3,4]*

5) What is *type((1))***?**

C. *int*

🧩 Python – Set — 5 MCQs

1) **How to create an empty set?**

B . set()

2). Which operation removes an element without raising if missing?

B. *discard*

3) What is *A ^ B* **for sets A and B?**

D. Symmetric difference

4) Which is TRUE about sets?

C. Membership check is average O(1)

5) What is the difference between *set* **and** *frozenset***?**

B. *frozenset* is immutable and hashable

🗂 Python – Dictionary — 5 MCQs

1) What does *in* **check for a dict** *d***?**

B. Keys

2) How to safely access a missing key with default?

B. *d.get('k', default)*

3) Dicts preserve insertion order since:

B. Python 3.6 (CPython impl detail) / guaranteed 3.7+

4) Merge two dicts *a* **and** *b* **(Python 3.9+)?**

C. a | b

5) Which can be a dict key?
D. *tuple* of hashables

---

pandas — 5 MCQs

1) Difference between *.loc* **and** *.iloc***?**

B. *.loc* is label-based; *.iloc* is integer-position based

2) Default join type in *pd.merge(left, right, on='key')***?**

C. Inner

3) Which converts a long format to wide format?

B. *pivot* / *pivot_table*

4) Which is best to apply different transforms to numeric vs categorical columns before modeling (sklearn)?

C. *sklearn.compose.ColumnTransformer*

5) Which reduces memory for repeated string-like categories?

B. *astype('category')*

Seaborn — 5 MCQs

1) Which Seaborn call sets theme globally (modern API)?

B. *sns.set_theme()*

2) Which is a *figure-level* function (returns *FacetGrid***)?**

C. *sns.relplot*

Explanation: Figure-level: *relplot*, *catplot*, *displot*, *lmplot*.

3. **Which parameter creates color categories?**

B. *hue*

4) Which statement about *sns.distplot* **is TRUE?**

B. It's deprecated; use *displot*/*histplot*/*kdeplot*

5) What does *sns.load_dataset("tips")* **do?**

B. Downloads example dataset from online repository

🤘 scikit-learn — 10 MCQs

1)Which methods characterize the estimator interface?

A. *fit*, *predict*, *transform* (where applicable)

2) Why use *Pipeline***?**

B. To chain preprocessing + model with correct CV behavior

3) Proper scaling practice when using *StandardScaler***?**

C. Fit only on train; transform train and test

4) Best split for imbalanced classification?

B. *train_test_split(..., stratify=y)*

5) What does *GridSearchCV* **do?**

B. Exhaustively searches specified grid with CV

6) Which metric is typically NOT for regression?

D. F1-score

7) Handling categorical variables in pipelines safely?
A. *OneHotEncoder(handle_unknown='ignore')*

8) Apply different preprocessing per column types?

B. *ColumnTransformer*

9) What does *random_state* **control?**

B. Reproducibility of stochastic operations

10) For SVMs and KNNs, feature scaling is:

C. Recommended/Important

🧠 TensorFlow / Keras — 10 MCQs

1) *Sequential* **vs Functional API?**

B. Functional supports arbitrary DAGs and multi-input/output

2) Proper setup for binary classification output layer & loss?

B. *Dense(1, sigmoid)*, *binary_crossentropy*

3) *sparse_categorical_crossentropy* **is appropriate when:**

C. Labels are integer class indices

4) Which returns a history of losses/metrics per epoch?

B. *model.fit*

5) What does *EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True)* **do?**

B. Stops if no improvement for 3 epochs and restores best weights

6) For input pipelines, which improves performance?
A. *dataset.prefetch(tf.data.AUTOTUNE)*

7) Which layer helps internal covariate shift and can stabilize training?

B. *BatchNormalization*

8) How to save a full Keras model (architecture+weights+optimizer state) in a portable way?

A. model.save('path')

9) Custom training loop uses:

B. *tf.GradientTape()*

10) For sequence models with padded inputs, which helps ignore padding?

A. *Masking* layer or *mask_zero=True* in *Embedding*

🔗 LangChain / LLM — 5 MCQs

1) What does *PromptTemplate* provide?

B. Structured prompts with variables/placeholders

2) What is an *LLMChain*?

B. A chain that combines a prompt template with an LLM to produce outputs

3) Purpose of *Memory* (e.g., *ConversationBufferMemory*) in LangChain?

B. Maintain conversational context across turns

4) What does *RetrievalQA* typically combine?

B. LLM + Retriever (vector store) for context-augmented answers

5) Which component indexes and searches chunks via embeddings most directly?

C. *VectorStore* (e.g., FAISS/Chroma)